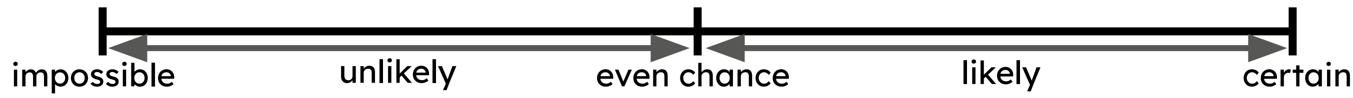
Name: ______________________ **Worksheet**

The probability scale

Task A: Expressing probabilities: percentages and decimals

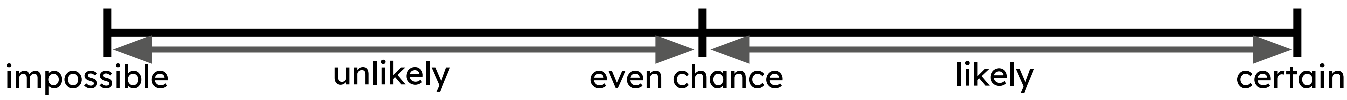
1) Write the **probabilities** below in the correct order on the **probability** scale.

0%, 100%, 20%, 75%, 50%, 45%, 55%

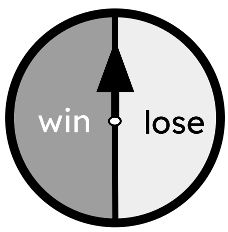


2) Write the **probabilities** below in the correct order on the **probability** scale.

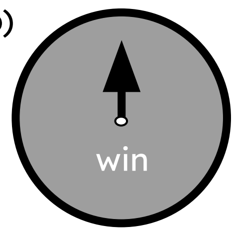
1, 0, 0.5, 0.1, 0.35, 0.95, 0.6



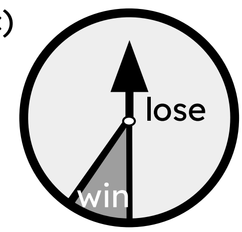
3) In each question you are given a spinner and a choice of four **probabilities**.
In each question, tick the **probability** that the spinner would land on 'win'.

a)


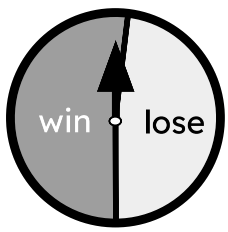
0.5%
1%
2%
50%

b)


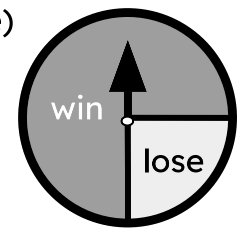
0
0.5
✓
100

c)


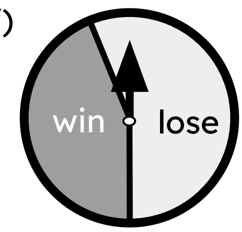
0.1
0.45
0.5
0.9

d)


40%
50%
52%
95%

e)


50%
52%
75%
98%

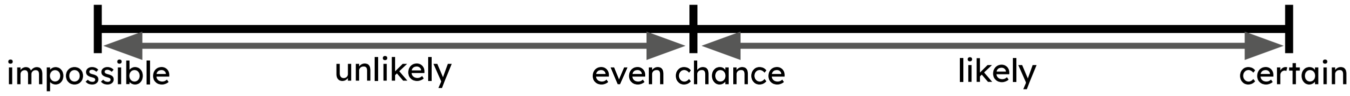
f)


0.1
0.45
0.5
0.6

Task B: Expressing probabilities as fractions

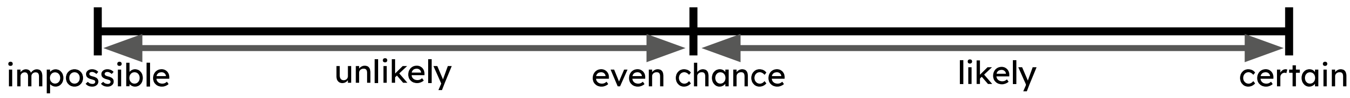
1) Write the **probabilities** below in the correct order on the **probability** scale.

$$\frac{20}{20} \quad \frac{1}{20} \quad \frac{5}{20} \quad \frac{18}{20} \quad \frac{11}{20} \quad \frac{9}{20}$$



2) Write the **probabilities** below in the correct order on the **probability** scale.

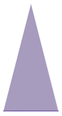
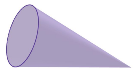
$$\frac{5}{6} \quad \frac{5}{5} \quad \frac{5}{1000} \quad \frac{5}{10} \quad \frac{5}{11} \quad \frac{5}{9}$$



3) Fill in the blanks in the table.

event	**probability as a fraction**	**probability as a decimal**	**probability as a percentage**	**description of likelihood**	**expected frequency in 600 trials**
example	$\frac{1}{2}$	0.5	50%	even chance	300
A		0.25		unlikely	150
B			75%		
C	$\frac{2}{5}$				
D					0

4) A dropped cone could land in one of two ways: ‘base down’ or ‘curve down’.
A large experiment concluded the following **probabilities**:

 P(base down) = $\frac{3}{10}$  P(curve down) = $\frac{7}{10}$

a) The cone is about to be dropped again. Which outcome is most likely to happen?

b) In an experiment with 100 trials, approximately how many times could you expect the cone to land curve down?

c) Use the link to access a simulation of this experiment: Cone simulation
The box that says ‘count’ shows the number of trials completed.
Press the ‘run’ button to begin the simulation and let it run for 100 trials.

Are the results the same as what you predicted in part (b)?